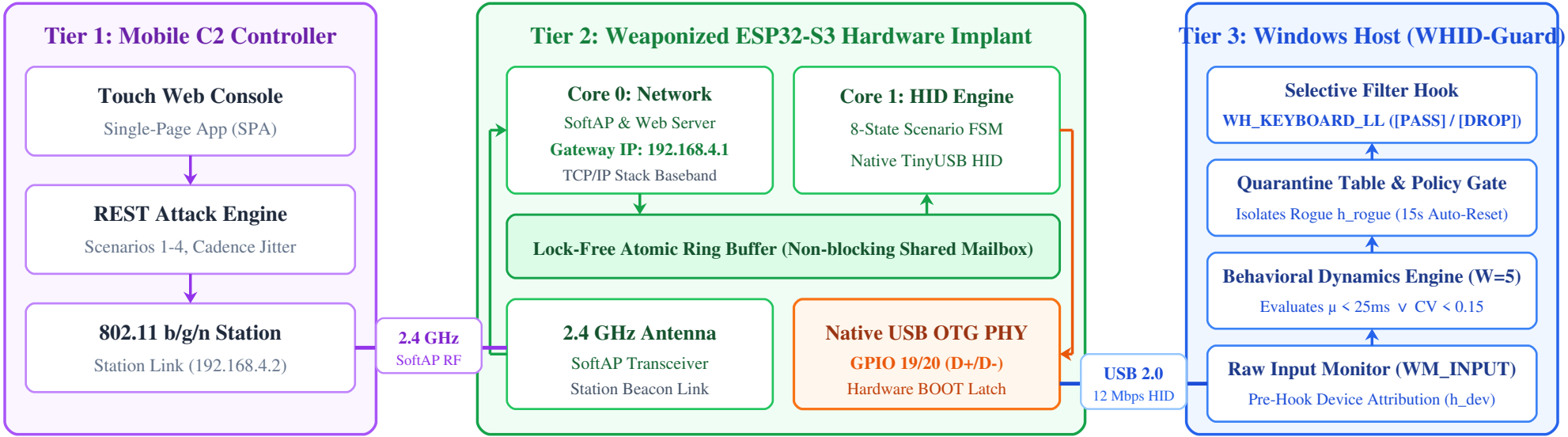

Tier 1: Mobile C2 Controller
Touch Web Console
Single-Page App (SPA)
REST Attack Engine
Scenarios 1-4, Cadence Jitter
802.11 b/g/n Station
Station Link (192.168.4.2)
2.4 GHz
SoftAP RF
Tier 2: Weaponized ESP32-S3 Hardware Implant
Core 0: Network
SoftAP & Web Server
Gateway IP: 192.168.4.1
TCP/IP Stack Baseband
Core 1: HID Engine
8-State Scenario FSM
Native TinyUSB HID
Lock-Free Atomic Ring Buffer (Non-blocking Shared Mailbox)
2.4 GHz Antenna
SoftAP Transceiver
Station Beacon Link
Native USB OTG PHY
GPIO 19/20 (D+/D-)
Hardware BOOT Latch
USB 2.0
12 Mbps HID
Tier 3: Windows Host (WHID-Guard)
Selective Filter Hook
WH_KEYBOARD_LL ([PASS] / [DROP])
Quarantine Table & Policy Gate
Isolates Rogue h_rogue (15s Auto-Reset)
Behavioral Dynamics Engine (W=5)
Evaluates μ < 25ms ∨ CV < 0.15
Raw Input Monitor (WM_INPUT)
Pre-Hook Device Attribution (h_dev)